

Effects of Melissa officinalis L. on oxidative status and DNA damage in subjects exposed to long-term low-dose ionizing radiation.

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The aim of this study was to determine the capability of *Melissa officinalis* L. (Lemon balm) infusion on improvement of oxidative stress status in radiology staff that were exposed to persistent low-dose radiation during work. The study was a before-after clinical trial performed on 55 radiology staff. They were asked to drink Lemon balm infusion which was prepared like a tea bag twice daily (1.5 g/100 mL) for 30 days. In the plasma, lipid peroxidation, DNA damage, catalase, superoxide dismutase, myeloperoxidase, and glutathione peroxidase activity were measured before and after using Lemon balm infusion. Use of Lemon balm infusion in radiology unit workers resulted in a significant improvement in plasma levels of catalase, superoxide dismutase, and glutathione peroxidase and a marked reduction in plasma DNA damage, myeloperoxidase, and lipid peroxidation. It is concluded that infusion of Lemon balm markedly improve oxidative stress condition and DNA damage in radiology staff when used as a dietary supplement for radiation protection.